This equation could further be written as,

$$I = \sum m\left(x^2 + d^2 + 2xd\right)$$

$$I = \sum\left(mx^2 + md^2 + 2dmx\right)$$

$$I = \sum mx^2 + \sum md^2 + 2d\sum mx$$

Here, $\sum mx^2$ is the moment of inertia of the body about the centre of mass. Hence, $I_C = \sum mx^2$

The term, $\sum mx = 0$ because, x can take positive and negative values with respect to the axis AB. The summation $\left(\sum mx\right)$ will be zero.

Thus, $I = I_C + \sum md^2 = I_C + \left(\sum m\right)d^2$

Here, $\sum m$ is the entire mass M of the object $\left(\sum m = M\right)$

$$I = I_C + Md^2$$

Hence, the parallel axis theorem is proved.

**(ii) Perpendicular axis theorem:**

This perpendicular axis theorem holds good only for plane laminar objects.

*The theorem states that the moment of inertia of a plane laminar body about an axis perpendicular to its plane is equal to the sum of moments of inertia about two perpendicular axes lying in the plane of the body such that all the three axes are mutually perpendicular and have a common point.*

Let the X and Y-axes lie in the plane and Z-axis perpendicular to the plane of the laminar object. If the moments of inertia of the body about X and Y-axes are $I_X$ and $I_Y$ respectively and $I_Z$ is the moment of inertia about Z-axis, then the perpendicular axis theorem could be expressed as,

$$I_Z = I_X + I_Y \qquad (5.47)$$

To prove this theorem, let us consider a plane laminar object of negligible thickness on which lies the origin (O). The X and Y-axes lie on the plane and Z-axis is perpendicular to it as shown in Figure 5.26. The lamina is considered to be made up of a large number of particles of mass m. Let us choose one such particle at a point P which has coordinates (x, y) at a distance r from O.

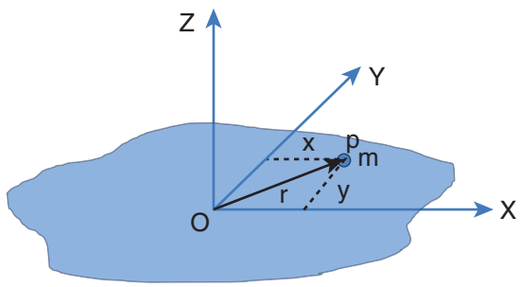



**Figure 5.26** Perpendicular axis theorem

The moment of inertia of the particle about Z-axis is, $mr^2$

The summation of the above expression gives the moment of inertia of the entire lamina about Z-axis as, $I_Z = \sum mr^2$

Here, $r^2 = x^2 + y^2$

Then, $I_Z = \sum m\left(x^2 + y^2\right)$

$$I_Z = \sum mx^2 + \sum my^2$$

In the above expression, the term $\sum mx^2$ is the moment of inertia of the body about the



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